



AERONAUTICAL
ENGINEERING



PROJECTS
PORTFOLIO

GPS-denied navigation · VIO/SLAM · EKF/UKF · GNC · Embedded ·
Reproducibility · Rocketry · Aeromodeling · Simulation · Fault tolerance

JUAN E. BERON

Sensor Fusion for Inertial Navigation System — National Central University (Taiwan)

NSTC | International Internship
Pilot Program

**TAIWAN EXPERIENCE
EDUCATION PROGRAM**



Date:

September 2025 – Present

Location:

Taoyuan, Taiwan

Sponsors:

NSTC International Internship Pilot Program, National Central University and TEEP (Taiwan Engineering Exchange Program).

Institution:

National Central University (NCU) — SenDiMoniProg

Inspector or Principal Responsible:

- Prof. Min-Chun Pan, Ph.D. — National Central University
- Jheng-Ruei Yu — Ph.D. Candidate, SenDiMonoProg Lab

Role:

Research Intern / Embedded Systems & Sensor Fusion Developer

Lead developer of the ROS 2 inertial sensing and calibration platform

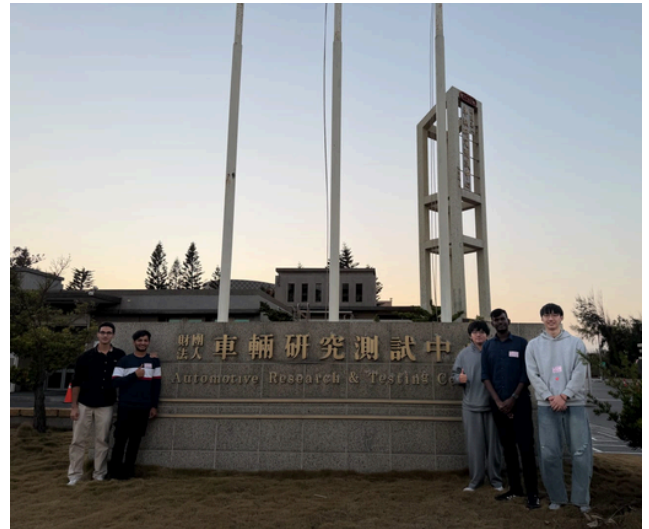
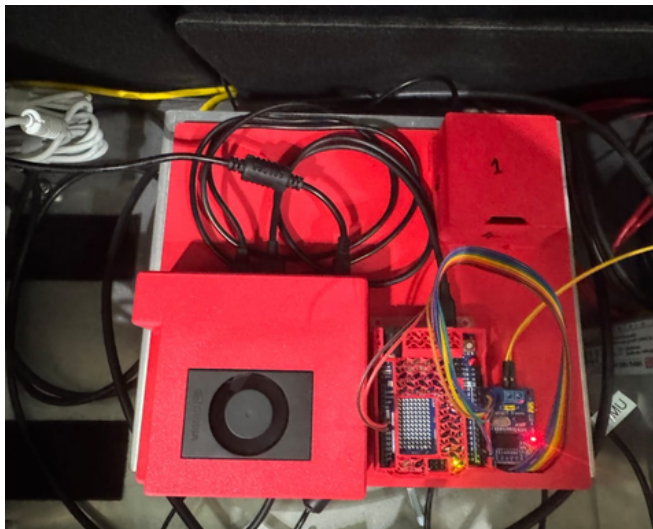


Figure 1 — Left: integrated IMU/GPS test setup prepared for ARTC validation; right: project team members at the Inertial Navigation System test site.

Summary

Development of a ROS 2-based inertial navigation platform intended as the foundation for multi-sensor fusion and SLAM in GPS-denied environments. The project focuses on building a reproducible, embedded, and calibration-aware architecture integrating IMU hardware, wireless communication, real-time visualization, and middleware tuning on NVIDIA Jetson. The resulting system serves as the inertial backbone for future integration of GPS, cameras, and SLAM-based tracking.

Problem

Advanced autonomy and SLAM systems depend on accurate, well-calibrated inertial data. In practice, sensor bias, misalignment, timing errors, and unreliable middleware configurations often dominate system error, especially in GPS-denied scenarios. Without a robust inertial platform that supports calibration, validation, and reproducibility, higher-level navigation and SLAM algorithms cannot be reliably evaluated or deployed

Objective

To design and implement a ROS 2-native inertial sensing and calibration platform capable of:

- Reliable high-rate IMU acquisition on embedded hardware
- Supporting online calibration and bias compensation for inertial sensors
- Providing low-latency, stable wireless communication for distributed autonomy systems
- Serving as a validated inertial reference for future IMU-GPS fusion and SLAM-based tracking

Detailed Project Description

The project was conceived as a system-level platform, not a single algorithm.

At the hardware level, a 9-DOF IMU was interfaced through an embedded pipeline and deployed on an NVIDIA Jetson Nano, selected after comparative evaluation against Raspberry Pi platforms. The Jetson provided sufficient computational margin for real-time ROS 2 workloads, visualization, and future GPU-accelerated SLAM and perception tasks.

On the software side, a containerized ROS 2 (Humble/Kilted) environment was developed using Docker. Core sensing functionality was implemented through custom ROS 2 nodes responsible for:

- IMU data acquisition and timestamp synchronization
- Publication of standardized `sensor_msgs/msg/Imu` topics
- Structured CSV logging for offline calibration and analysis

A quaternion-based AHRS pipeline (Madgwick/Mahony) was implemented to provide real-time attitude estimation at ~200 Hz. Significant emphasis was placed on calibration and signal conditioning, including:

- Dynamic gyroscope and magnetometer bias compensation
- Low-pass Butterworth pre-filtering (5 Hz)
- Complementary fusion to mitigate angular drift

Wireless communication between the Jetson and a remote GUI was optimized by migrating to a custom UDPv4 Fast DDS transport, defined through tailored XML QoS profiles. Network behavior was systematically profiled under different configurations, enabling reduction of latency, jitter, and packet loss. For observability and validation, a real-time visualization and debugging stack was developed using RViz2, tf2, rqt_graph, and ros2 bag tools. The system supports simultaneous real-time monitoring and synchronized logging, enabling both live diagnostics and post-processing.

The platform is currently being extended toward:

- IMU–GPS calibration and EKF-based sensor fusion
- Inertial support for SLAM pipelines (e.g., `slam_toolbox`, Nav2)
- Integration with vision sensors for visual-inertial navigation

Skills Involved

1. Inertial navigation & estimation: AHRS, bias estimation, noise modeling
2. Sensor fusion: quaternion representations, reference frames, EKF readiness
3. Autonomous systems: ROS 2 architecture, Nav2, SLAM integration planning
4. Embedded systems: NVIDIA Jetson Nano, real-time constraints
5. Middleware & networking: Fast DDS, QoS tuning, wireless latency profiling
6. Software engineering: Docker, reproducibility, GitHub-based research workflows
7. Research methodology: experimental validation, metrics, failure-mode analysis

Results and Key Outcomes

- Real-time IMU pipeline (ARTC validation): Deployed and tested the ROS 2 (Humble/Kilted) IMU stack on Jetson Nano during trials at ARTC, sustaining ~195–200 Hz publishing with synchronized timestamping and logging.
- Low-latency wireless transport: After migrating to a custom UDPv4 Fast DDS transport and profiling (RTT/packet loss), average latency dropped from ~170 ms (non-dedicated network) to sub-40 ms, with sub-10 ms end-to-end ROS 2 message delay in the dedicated setup.
- Calibrated attitude estimation: Implemented AHRS (Madgwick/Mahony) at ~200 Hz with online bias compensation + 5 Hz Butterworth filtering, achieving ~0.04 rad/s RMS orientation-estimation error under lab conditions.
- Jitter reduction: Transport/QoS tuning reduced network jitter by $\approx 38\%$ versus baseline profiles.

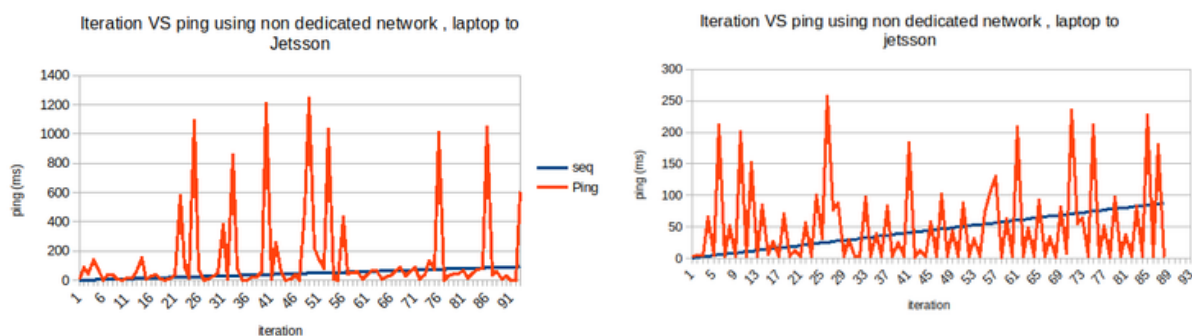


Figure 2 — Left: end-to-end network latency and jitter for the baseline DDS configuration on a shared IoT network; right: reduced latency after custom Fast DDS transport and dedicated port assignment for each subsystem.

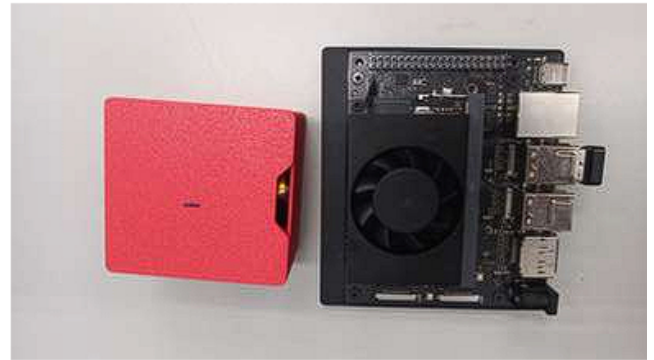
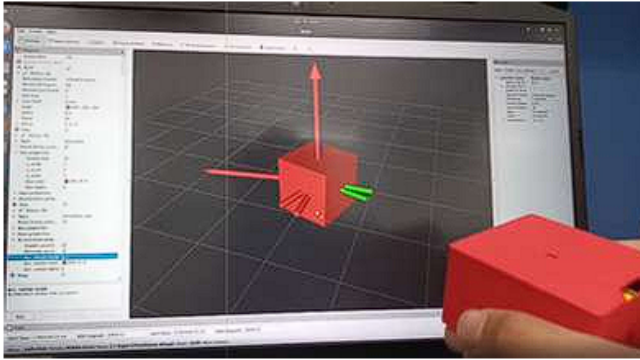


Figure 3 — Left: IMU pose and inertial behavior visualized in ROS2 and RViz after integration; right: ROS2 Dockerized setup combining the inertial platform with a depth camera for multi-sensor testing.

Evidence

1. Demonstration Video : live test integrating ROS 2 with IMU acquisition through an ESP32 and Jetson Nano, publishing sensor topics wirelessly and visualized in RViz2. ([Link](#))
2. Project repo containing Dockerized ROS 2 packages, launch files, and simulation nodes for multi sensor testing and algorithm validation used during this research. ([Link](#))

CUBESAT PAYLOAD — Multisensor Atmospheric Probe | Universidad de Antioquia & UPB | 2024–2025



Date

April 2024 – June 2025

Location

Medellín, Colombia

Institution

Universidad de Antioquia

Inspector / Principal Responsible

- Dr. Francisco Hidalgo — Professor of Aerodynamics
- Volta Rocketry — External technical supervision

Role

Responsible for descent dynamics modeling, sensor integration, and state reconstruction

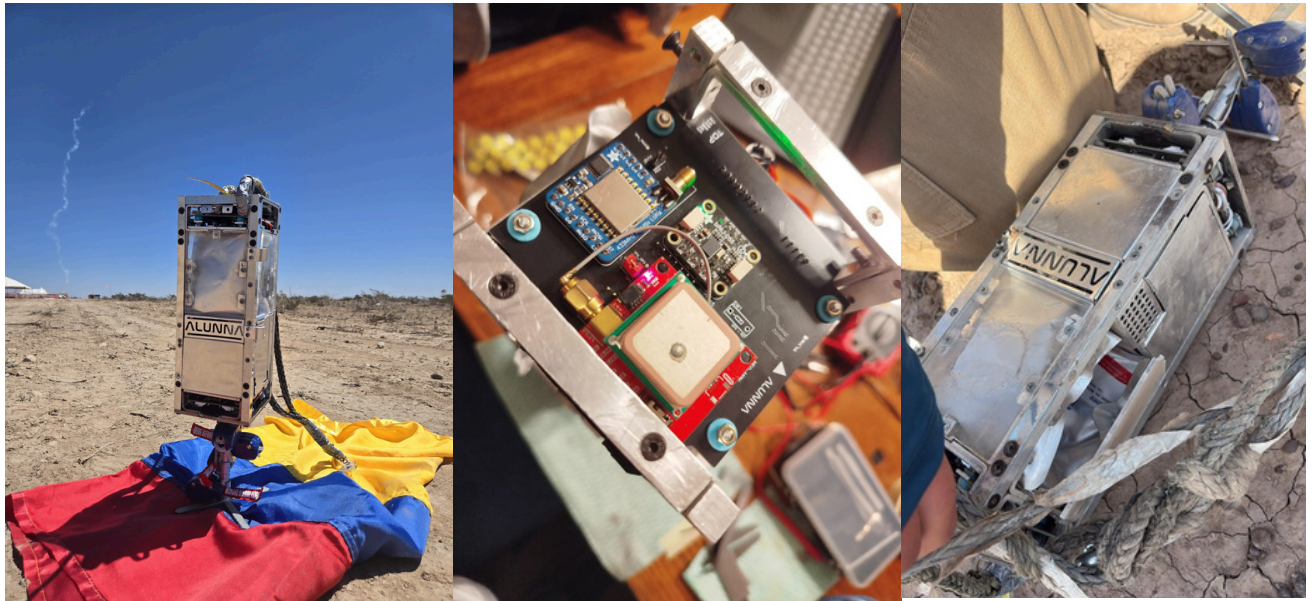


Figure 1 — ALUNA CubeSat Flight and Recovery: left: ascent phase during Spaceport America Cup 2025 prior to CubeSat deployment, middle: integrated multisensor payload (IMU, GPS, Pitot system) inside the CubeSat, right: recovered CubeSat structure after atmospheric descent showing impact condition and sensor-module survivability.

Summary

Design and analysis of a CubeSat atmospheric descent experiment focused on dynamic modeling, multi-sensor state estimation, and post-flight reconstruction. The project integrated IMU, GPS, and multi-Pitot pressure sensing to estimate velocity, attitude, and aerodynamic behavior during descent. Analytical modeling, CFD simulations, and experimental data were combined to validate system performance and reconstruct the vehicle trajectory.

Problem

Atmospheric descent experiments in small satellites and sounding payloads are constrained by limited sensing, harsh dynamics, and short data windows. Accurate reconstruction of velocity, attitude, and aerodynamic coefficients requires robust sensor integration and physically consistent models, especially when GPS resolution or availability is limited and pressure measurements are noisy.

Objective

To develop and validate a CubeSat-scale descent estimation framework capable of:

- Modeling atmospheric descent dynamics under realistic assumptions
- Estimating velocity and attitude using IMU, GPS, and multi-Pitot data
- Reconstructing the post-flight trajectory through sensor fusion and physical modeling
- Comparing experimental results against CFD-based aerodynamic predictions

Detailed Project Description

The project began with the formulation of a 6-DOF descent dynamics model, accounting for gravity, drag, and atmospheric density variations. A CubeSat-class payload was instrumented with:

- IMU for angular rates and accelerations
- GPS for coarse position and velocity reference
- Multi-Pitot tube array for dynamic pressure and airflow estimation

Sensor data were logged during descent and processed offline. IMU measurements were used to reconstruct attitude evolution, while Pitot pressure readings were converted into airspeed estimates using calibrated flow models. GPS data served as a reference for validation and drift correction. To support aerodynamic modeling, the vehicle geometry was analyzed using CFD simulations, estimating drag coefficients and pressure distributions across relevant Reynolds number regimes. These coefficients were incorporated into the descent model and compared against experimentally reconstructed values.

The post-flight reconstruction combined sensor data and dynamic equations to estimate:

- Velocity profiles
- Attitude evolution
- Aerodynamic forces during descent

Discrepancies between modeled and measured behavior were analyzed to identify sensor limitations, flow separation effects, and modeling assumptions.

Skills Involved

- Flight dynamics: 6-DOF descent modeling, drag and load estimation
- State estimation: IMU/GPS fusion, trajectory and attitude reconstruction
- Aerodynamics: Pitot-tube calibration, wind-tunnel testing, CFD-based drag analysis
- Data analysis: signal filtering, RMS error metrics, model–experiment comparison
- Experimental methods: flight-data logging, sensor mounting, post-flight reconstruction
- Scientific computing: Python/MATLAB for processing, visualization and validation

Results and Key Outcomes

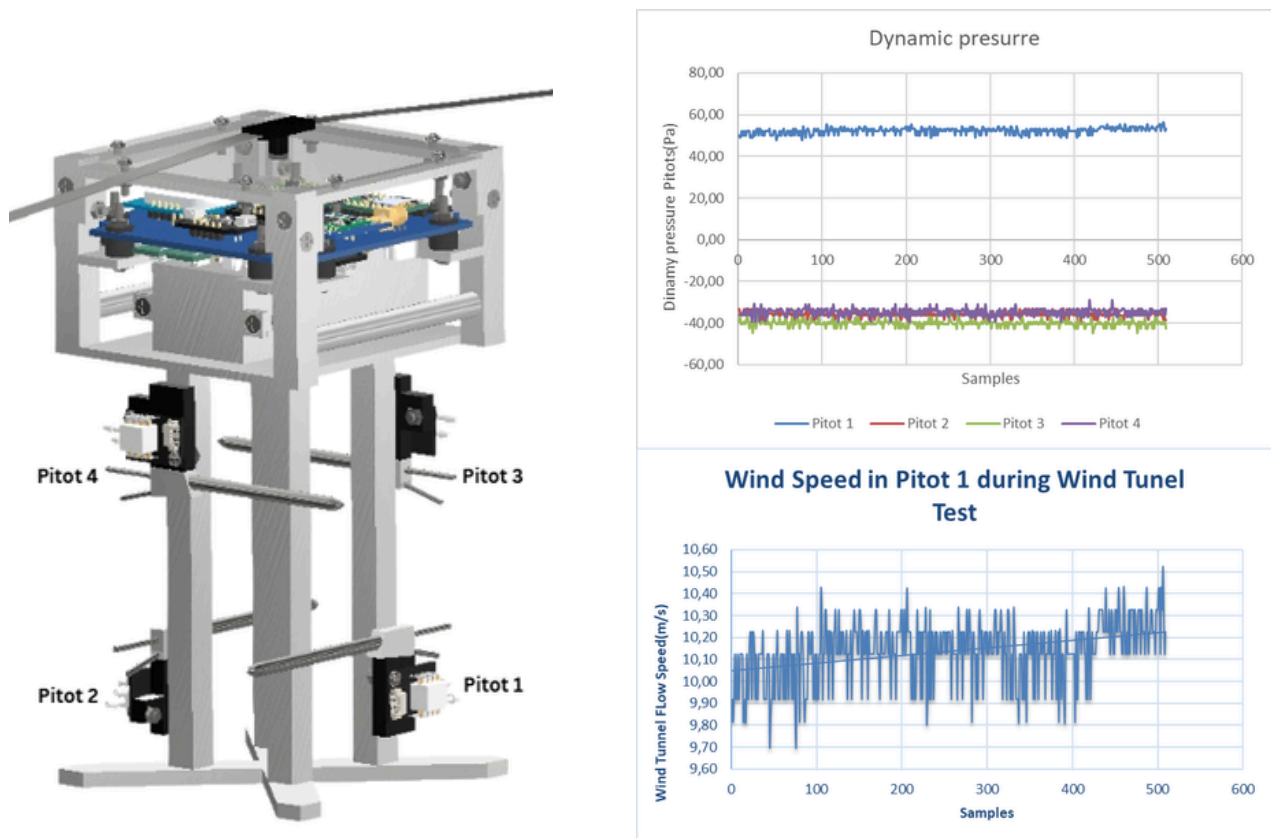


Figure 2 — Pitot System CAD Model and Wind-Tunnel Calibration Data: left: full CAD model of the CubeSat with the assigned positions of Pitots 1–4, top-right: wind-tunnel dynamic-pressure signals from Pitots 1–4 showing directional response and pressure stability, bottom-right: airspeed reconstruction from Pitot 1 aligned parallel to the wind-stream during calibration.

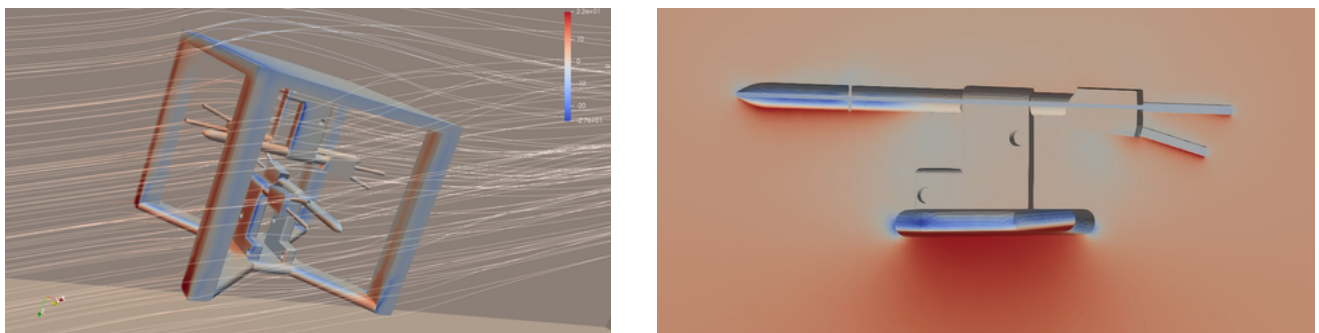


Figure 3 — CFD Aerodynamic Simulations: left: external airflow streamline simulation evaluating structural influence on aerodynamic behavior, right: internal CFD flow visualization showing pressure development inside the Pitot ducts.

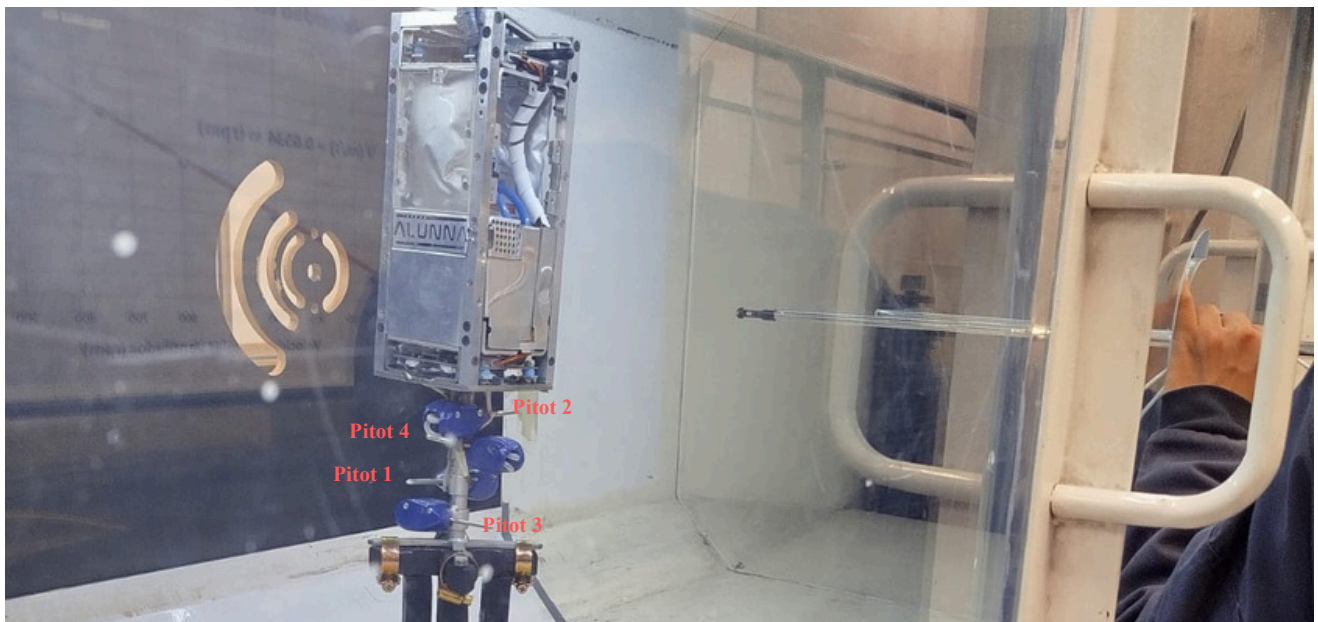


Figure 4 — CubeSat Wind Tunnel Assembly: full Pitot-array hardware mounted in the wind tunnel for aerodynamic validation and directional calibration, showing the orientation of Pitots 1–4 relative to the freestream.

Results and Key Outcomes

- Flight deployment and data capture: Deployed the ALUNNA CubeSat at approximately 10 km altitude during the Spaceport America Cup ascent and recovered the payload with ≈ 30 s of usable IMU, GPS and Pitot data from the descent phase.
- Pitot array calibration: Wind-tunnel tests showed Pitot 1 dynamic pressure around 50–55 Pa with small fluctuations, while Pitots 2–4 remained near zero or slightly negative, confirming clear directional sensitivity and correct mounting geometry.
- Airspeed estimation performance: The calibrated Pitot-1 signal produced a wind-speed trace clustered around ≈ 10 m/s over ~ 600 samples, with high-frequency variations matching tunnel turbulence, validating the pressure–velocity conversion model used for flight reconstruction.
- CFD–experiment agreement: CFD streamlines and pressure fields reproduced the stagnation and wake patterns seen in tunnel testing, yielding aerodynamic behavior consistent with measured Pitot gradients and supporting the drag model for descent simulations.
- Improved descent reconstruction: Fusing Pitot-based airspeed with IMU and GPS data reduced reliance on sparse GPS samples and limited IMU drift, enabling a physically consistent reconstruction of descent velocity and aerodynamic loads for the post-deployment trajectory.

More Evidence:

- Volta Rocketry — Satellite Projects: Overview of ALUNNA and related CubeSat/CanSat efforts: objectives, architecture, integration photos, and test reports. ([Link](#))
- Git-Hub Repository used in this project ([Link](#))

SPACEPORT AMERICA CUP / IREC — Volta Rocketry (UdeA) — Payload & Telemetry | 2025

**Date:**

April 2024- June 2025

Location:

Medellin, Colombia

Sponsors:

Universidad de Antioquia, ALAS, HATCH

Institution:

Universidad de Antioquia

Inspector or Principal Responsible:

- Dr. Francisco Hidalgo — Professor of Aerodynamics
- Volta Rocketry — External technical supervision

Role:

Payload & Telemetry Engineer

Responsible for sensor integration, payload–airframe interfaces, flight-data acquisition, and post-flight telemetry reconstruction.



Figure 1 — Rocket Subsystems Team: group photo with the scale model and development airframe used for payload integration and subsystem coordination.

Summary

Design, integration, and testing of the payload (CanSat) and telemetry subsystem for the Volta Rocketry 10k-ft COTS rocket competing at the Spaceport America Cup / IREC 2025. Work included sensor acquisition, electrical integration, environmental testing, modeling expected flight loads, and handling data operations during launch day. The payload successfully acquired flight data during ascent and supported the recovery and post-flight analysis pipeline.

Problem

Payloads in collegiate high-power rocketry experience high acceleration, vibration, thermal gradients, short measurement windows, and strict mass/volume limits. A telemetry-capable payload must maintain structural integrity, record reliable measurements through the ascent profile, and interface safely with the recovery system—all while meeting competition regulations.

Objective

To design and implement a robust payload and telemetry system capable of:

- Withstanding launch-induced acceleration, vibration, and temperature variations
- Acquiring usable flight data (acceleration, orientation, temperature, altitude)
- Integrating mechanically and electrically with the rocket's avionics and recovery system
- Supporting post-flight data curation and preliminary trajectory reconstruction

Detailed Project Description

The project began with defining environmental conditions for the 10k-ft COTS category: expected accelerations, motor thrust curves, aerodynamic loading, and internal airframe constraints. Based on these parameters, the payload was designed as a CanSat-class module containing:

- IMU + barometric sensing for acceleration and altitude profiling
- Telemetry and onboard logging circuitry
- Structural interfaces for mounting within the rocket airframe
- Integration with separation and recovery systems

Testing included bench-level diagnostics, environmental checks (vibration + handling), and simulations of the expected ascent loads. During competition week, the payload underwent:

- Pad-integration rehearsals and final checks
- Launch-day data operations and telemetry readiness verification
- Post-flight data retrieval and curation for analysis

The team executed a successful launch at Spaceport America Cup 2025, achieving the target category's flight profile and validating the payload's structural and electrical integrity.

Skills Involved

- Launch-vehicle environments: accel/temperature/structural constraints
- Experimental payload design & structural integration
- Telemetry and data logging
- Basic flight-dynamics prediction for payload conditions
- Recovery-system interface design
- Launch-day operations, diagnostics, and post-flight data handling



Fig.2 – Rocket Subsystems Team — Group Photo with Scale Model and Development Airframe (Volta Rocketry)

Results and Key Outcomes (Metrics-Driven)

1. Successful competition launch: The rocket achieved the expected ascent profile for the 10k-ft COTS category, with clean deployment and structural integrity maintained for payload recovery.
2. Payload survivability and data capture: The payload preserved onboard data despite high acceleration and thermal conditions, yielding a complete ascent telemetry set suitable for post-flight analysis.
3. Directional acceleration and thermal trends: Retrieved data showed acceleration patterns consistent with motor thrust curve predictions, and temperature readings remained within operational limits throughout ascent and recovery.
4. Integration reliability: Mechanical and electrical interfaces remained secure during liftoff, max-Q, and deployment, confirming the robustness of the airframe–payload mounting system.
5. Team performance: Volta Rocketry achieved Top-5 in Design & Build Quality, Top-7 in Technical Report, and Top-15 overall in the 10k COTS category, demonstrating strong technical standing among international teams.

More Evidence:

- ESRA — Official Results (2025): Fly-off rankings, category standings (10k COTS), and scoring summaries for SA Cup 2025. ([Link](#))
- Volta Rocketry — Official Website: Central hub with project milestones, build logs, team roster, and current research initiatives ([Link](#))
- Volta Team Blog — Competition & Progress Updates: Event recaps, integration days, mission readiness reviews, and post-flight notes ([Link](#))

Conceptual Design of the Electrical and Spray Subsystem for an Agricultural Aircraft



Date

November 2025

Location

Medellín, Colombia

Institution

Universidad Pontificia Bolivariana (UPB) — Aeronautical Engineering Program

Inspector / Principal Responsible

Juan Pablo Alvarado — Professor of Aircrafts Design/ Aerodynamics

Role

Aeronautical Engineering Student - Final Task Aircraft Design Course

Summary

This project develops the electrical power system and chemical spraying subsystem for a high-productivity agricultural aircraft proposed under the AIAA “Safe High Operational Tempo Sprayer” (SHOTS) Flight Research Problem. The design combines a 28 VDC dual-source architecture (starter-generator and standby alternator), redundant LiFePO₄ batteries, and a 27 m spray boom with 108 flat-fan nozzles, a 700 L/min pump and GNSS/autopilot-based rate control. Hydraulic models, mission segmentation, and electrical load analysis are integrated to size generator power, battery endurance, wiring and protection, while ensuring spray rate, droplet spectrum, and power-quality margins that meet modern agricultural and aviation standards.

Problem

Modern agricultural aircraft must deliver high field throughput, precise chemical application and safe redundancy under demanding operating conditions. For the SHOTS problem, the aircraft must:

- Operate across 20–60 L/ha application rates and 50–80 m/s ground speeds.
- Treat large fields efficiently with a 27 m boom and large hopper volume.
- Maintain electrical power to critical loads after a generator failure for a defined time window.
- Guarantee acceptable voltage drops, current margins and spray quality (droplet size and boom uniformity).

Designing a system that simultaneously satisfies hydraulic, electrical, and operational constraints is the core challenge.

Objective

To design and validate an integrated 28 VDC electrical and agricultural spraying system capable of:

- Supplying adequate hydraulic power for the full SHOTS spray envelope.
- Providing sufficient installed generator capacity and battery energy with margin.
- Keeping voltage drops and current levels within aviation power-quality limits.
- Achieving uniform flow and pressure along a 27 m boom for accurate application.
- Meeting sortie-level coverage requirements for large agricultural fields.

Detailed Project Description

Spray system and hydraulic sizing

The spray subsystem is based on a 27 m wing-mounted boom with 108 flat-fan nozzles spaced at 0.25 m. Each nozzle operates around a nominal pressure of 350 kPa, with flow sufficient to cover application rates between 20 and 60 L/ha over ground speeds from 50 to 80 m/s. A 700 L/min pump is selected to cover the worst-case combination of high speed and high application rate, with margin relative to the required flow envelope.

Hydraulic models relate the application rate (a), ground speed (V_g), boom width (b) and total flow (Q), and from these variables the required hydraulic power ($Phyd$) is obtained using the relation $Phyd = \Delta p * Q / \eta$. Across the SHOTS operating envelope, the required hydraulic power varies from approximately 1.1 kW to 4.5 kW, with a design point of about 3.9 kW used to size both the pump and the electrical drive.

Electrical architecture and load analysis

The aircraft uses a 28 VDC architecture with:

- A main starter-generator rated around 8.4 kW (continuous),
- A standby alternator around 2.1 kW,
- Two 24 V / 40 Ah LiFePO₄ batteries feeding an essential bus (B ESS) and a main bus (B MAIN).

A detailed one-line diagram maps all loads (avionics, servos, spray pump, valves, lights, auxiliary systems) to each bus. The mission is segmented into ground, transit, spray, and return phases, and for each segment the current draw of both buses is computed. This allows the design of:

- Generator loading profiles,
- Battery-only endurance in case of generator failure,
- Shedding strategies to keep essential loads powered.

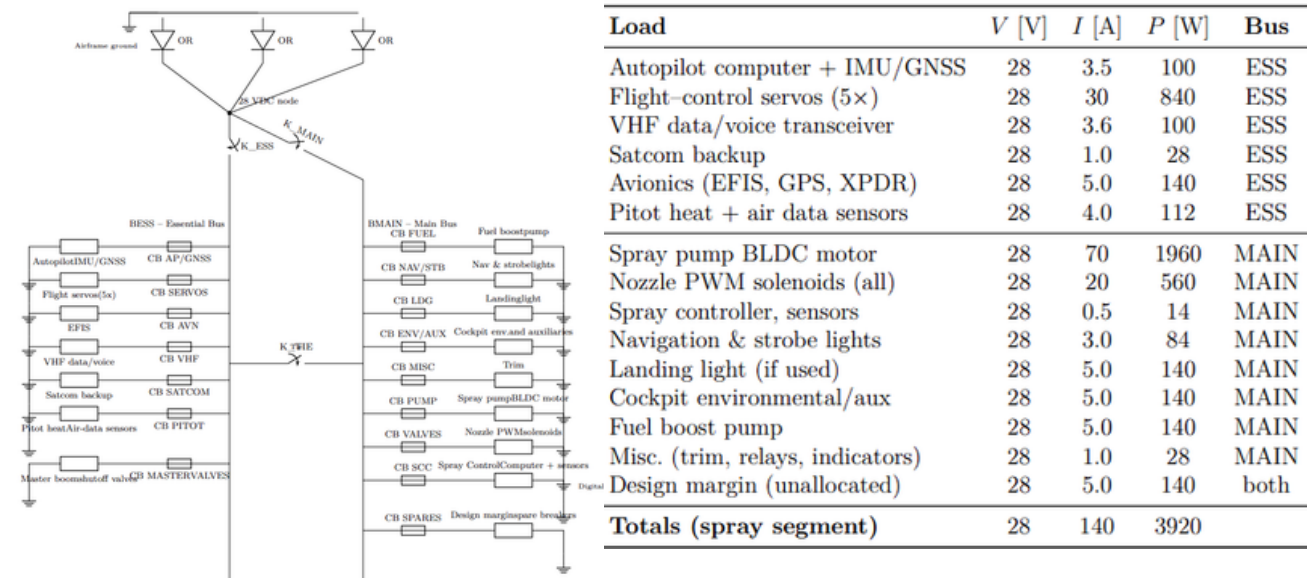


Figure 1 — Left: detailed one-line diagram of the proposed 28 VDC electrical and spray-power architecture for the SHOTS agricultural aircraft, designed to satisfy the Flight Research Problem requirements; right: load summary table listing each electrical and spraying subsystem with its nominal voltage, current and power demand, together with its allocation to the essential or main bus.

Power, energy and voltage-drop evaluation

Using the mission load profile, the average current and power for each segment are calculated. During spray, generator loading reaches about half of its continuous rating, while mission-averaged electrical power is ≈ 2.7 kW (≈ 97 A at 28 V), well below installed capacity. Battery endurance on the essential bus is evaluated against the requirement that the aircraft must remain safely powered for a defined time after loss of generation; the selected batteries provide roughly 40 minutes of ESS-only operation versus a 30-minute requirement, leaving a comfortable margin.

Cable gauges and line lengths are then used to compute voltage drops for worst-case currents. All main branches remain within roughly 2–5% voltage drop, consistent with aviation and industrial power-quality guidelines.

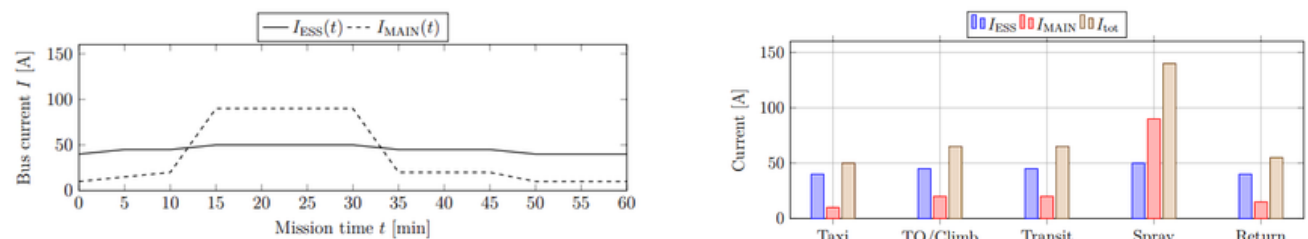


Figure 2 — Left: time history of current draw on the essential and main 28 VDC buses over the full mission, resolved by operating phase (taxi, takeoff, climb, transit, spray and return); right: average current carried by each bus in every flight segment, highlighting the relative loading of the essential versus main buses throughout the sortie.

Application quality and field throughput

The boom internal diameter and manifold layout are analyzed to ensure uniform pressure along the 27 m span, with larger-diameter plumbing selected to keep tip-to-root pressure variation within a narrow band, supporting consistent nozzle flow. Nozzle manufacturer data and pressure–flow curves are used to estimate droplet size spectra (VMD), targeting coarse/very-coarse droplets to reduce drift risk while maintaining agronomic efficacy.

Finally, the hopper volume (e.g., 3000 L) and selected application rates are combined to compute treated area per sortie. At 30 L/ha, the system can cover roughly 100 ha per load, and at 20 L/ha up to ≈ 150 ha, allowing a 400-acre field to be treated in about two sorties under nominal conditions.

Skills Involved

- Hydraulic modeling of agricultural spray systems (flow, pressure, power)
- Electrical power-system design (28 VDC architecture, buses, protection, shedding)
- Power and energy budgeting over complex missions
- Voltage-drop and conductor sizing analysis
- Nozzle selection and droplet-spectrum reasoning (VMD, application quality)
- Mission and operations analysis for high-throughput agriculture
- System-level integration of avionics, power and payload subsystems

Results and Key Outcomes

1. Hydraulic power envelope sized and covered: Modeling of the boom and pump shows that required hydraulic power remains between ≈ 1.1 and ≈ 4.5 kW across the SHOTS operating envelope (20–60 L/ha, 50–80 m/s). A design point around ≈ 3.9 kW was used to size the electrical pump drive, and the selected 700 L/min pump can meet all required flow conditions with margin.
2. Sufficient generator power with comfortable margin: The main ≈ 8.4 kW starter-generator and ≈ 2.1 kW standby alternator provide installed power above the combined electrical and hydraulic requirements. During spray, generator loading is around 50% of continuous rating, indicating ample thermal and reliability margin for sustained operations.
3. Mission-averaged loading well below capacity: Over a representative mission, average electrical loading is approximately 97 A at 28 V (≈ 2.7 kW), significantly lower than installed capacity. This ensures the system can accommodate transients, growth in auxiliary loads, and degradation without violating power limits.
4. Battery endurance exceeding minimum requirements: Analysis of the essential bus (B ESS) shows that the dual 24 V / 40 Ah LiFePO₄ battery configuration provides roughly 40 minutes of standalone endurance, exceeding the 30-minute post-failure requirement. This supports a safe return or controlled mission abort in case of generator loss.
5. Controlled voltage drops and power quality: Conductor sizing and routing yield worst-case voltage drops in the 2–5% range on critical branches, keeping equipment within acceptable supply limits and aligning with typical aviation and industrial 28 VDC standards.
6. High field coverage efficiency: With a 27 m boom and 108 nozzles, the system can treat approximately 100 ha per sortie at 30 L/ha, and ≈ 150 ha at 20 L/ha, enabling a 400-acre field to be sprayed in about two sorties under nominal conditions. This meets the high-throughput goals of the SHOTS concept while maintaining acceptable droplet size and boom-uniformity characteristics.

Metric	SHOTS	AT-802A	Large UAV
Hopper volume [L]	3000	~ 3000	100–600
Nom. rate [L/ha]	30	30	10–30
Spray speed [m/s]	70	60–70	10–25
Flow Q [L/min]	~ 340	300–400	20–150
Spray pump power [kW]	3–5 (electric)	2–4 (mech./hyd.)	1–3 (electric)
Main electrical gen. [kW]	8 (dual-source)	5–10	Battery only
Redundancy	Dual gen., dual bat., dual bus	Often single gen.	Typically single bus
Area/sortie @30 L/ha [ha]	100	~ 100	3–20

Figure 3 — Comparative summary of the principal geometric, payload and spray-performance characteristics of the proposed SHOTS agricultural aircraft relative to representative agricultural aircraft currently available on the market.

More Evidence:

- Design of the Electrical and Spray Subsystem for an Agricultural Aircraft A Conceptual Engineering Proposal under the AIAA Functional Requirements_ Final project for Aircrafts Design course ([Link](#))

UAV Simulation & Autonomy — ROS2 / Multi-Sensor Environments | 2023–2024 | Daetech Simulation Lab

Date

june 2023 - September 2024

Location

Medellín, Colombia

Institution

Daetech Simulation Lab — Universidad Pontificia Bolivariana

Inspector / Principal Responsible

Juan Pablo Alvarado – Aerospace Engineering Lab Coordinator

Role

Simulation & UTM Algorithm Engineer

Responsible for urban-corridor generation from satellite data, 3D city modeling, and ROS2/Gazebo integration for multi-UAV simulations.

Summary

Development of an urban air-mobility (UAM) simulation pipeline that transforms real satellite imagery of a city into a discrete airspace grid and a 3D virtual environment, and then uses that representation to simulate drone traffic in ROS2/Gazebo. The workflow converts Copernicus multispectral imagery into a pixel matrix of free and occupied cells, extracts aerial corridors above streets and open areas, reconstructs the city in Blender, and exports the scene to ROS2 so that UAVs can fly along the corridors using waypoints generated by the algorithm.

Problem

Testing concepts for urban air mobility and UTM (Unmanned Traffic Management) requires a realistic representation of the city, including buildings, streets, rivers and no-fly zones, as well as a way to encode this information as a discrete navigation space. Conventional UAV simulators either provide overly simplified maps or require manual modeling of the environment. There is a need for an automated pipeline that:

Derives air corridors and no-fly regions directly from Earth-observation data,

Encodes the city as a discrete grid suitable for path-planning algorithms, and

Links this grid to a 3D simulation where UAM concepts can be tested with multiple drones.

Objective

To design and implement a UAM/UTM simulation framework capable of:

Converting Copernicus satellite imagery of a city into a pixel-based occupancy grid,

Extracting candidate aerial corridors above streets, rivers and other suitable features,

Building a corresponding 3D city model in Blender aligned with the grid,

Exporting the environment to ROS2 + Gazebo and injecting the corridor coordinates as waypoints for multi-UAV simulations.

Detailed Project Description

Image processing and corridor extraction

Copernicus multispectral imagery of the target urban area is first pre-processed (cropping, georeferencing and band selection) and then segmented to distinguish built-up areas, streets, open spaces and restricted zones. The city is discretized into a 2D raster grid, where each pixel is classified as either occupied, no-fly or free corridor cell. Streets, rivers and open strips are used to define continuous chains of free cells that form aerial corridors and intersections, which are stored as a graph structure for planning.

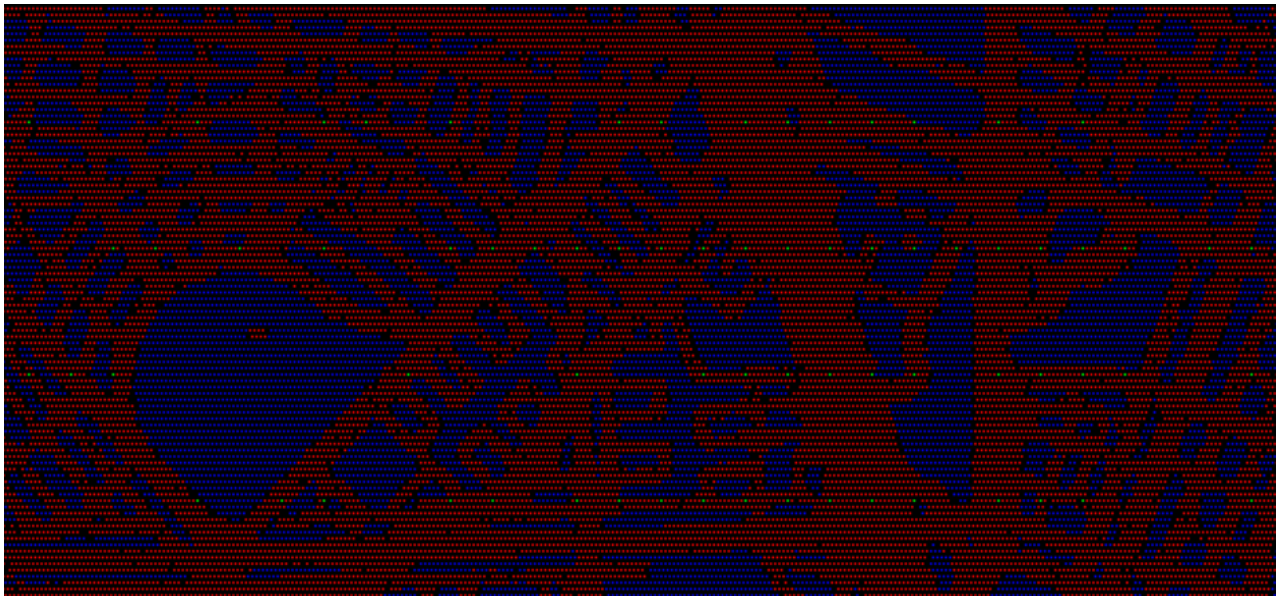


Figure 1 — Pixel-based corridor map generated from Copernicus imagery, used as the base discretized field for the algorithm's route and air-corridor construction.

Pixel-grid representation and waypoint generation

The corridor graph is converted into a matrix representation compatible with typical occupancy-grid and path-planning algorithms. Each free cell can be interpreted as a “hop” location for a UAV, and sequences of cells are transformed into 3D waypoints by assigning altitude layers above the ground. These waypoints define safe trajectories that respect the corridor structure and avoid no-fly regions derived from the imagery.

3D city reconstruction in Blender

Using the same spatial reference, the city is reconstructed in Blender: building footprints are extruded into volumes, streets and rivers are modeled as surface features, and basic lighting and atmospheric settings are added to improve visual realism. A Python script ensures that the Blender scene remains registered to the underlying grid, so that corridor cells correspond to the correct locations in the 3D model.

ROS2 and Gazebo integration

The Blender environment is exported as a Gazebo-compatible world (e.g., via COLLADA/FBX), and imported into a ROS2 + Gazebo simulation. UAV models are spawned in this environment and controlled by ROS2 nodes that consume the precomputed waypoint lists. The nodes publish setpoints derived from the corridor matrix, allowing drones to “hop” from cell to cell along aerial corridors while avoiding obstacles. Additional ROS2 components handle telemetry, logging and visualization, making the platform suitable for testing UTM strategies (conflict detection, sequencing, and rerouting).

Skills Involved

- Remote sensing and basic GIS (Copernicus imagery preprocessing and segmentation)
- Occupancy-grid and graph-based path-planning representations
- Blender + Python for 3D urban environment modeling
- ROS2 and Gazebo for UAV simulation and waypoint-based navigation
- Integration of perception, planning and control in a unified simulation pipeline
- Dockerized development and reproducible simulation setups (where applicable)

Results and Key Outcomes

1. End-to-end UAM simulation pipeline: Implemented a complete workflow from Copernicus imagery to ROS2/Gazebo simulations, enabling urban air-corridor experiments without manual city modeling.
2. Discrete airspace representation: Generated a pixel-based occupancy grid and corridor graph for a representative urban district, providing a reusable testbed for UTM path-planning and traffic-management algorithms.
3. 3D city and UAV integration: Successfully aligned the Blender-based 3D city model with the discrete grid, allowing UAVs in Gazebo to follow corridor waypoints that correspond to real urban features (streets, rivers and open areas).
4. Real-time simulation performance: Achieved stable real-time operation of the ROS2/Gazebo environment at 60 Hz, supporting multi-UAV scenarios and closed-loop testing of guidance and avoidance logic.
5. Foundation for UTM research: The framework now serves as a research platform for evaluating urban-air-mobility concepts, including corridor-based navigation, traffic-management policies and AI-driven autonomy in realistic city geometries.

More Evidences:

- Path builder Algorithm Repository ([Link](#))

AIAA DESIGN/BUILD/FLY (DBF) — Propulsion & Electrical Systems | 2021–2022 | UPB “DBF Medellín Team”

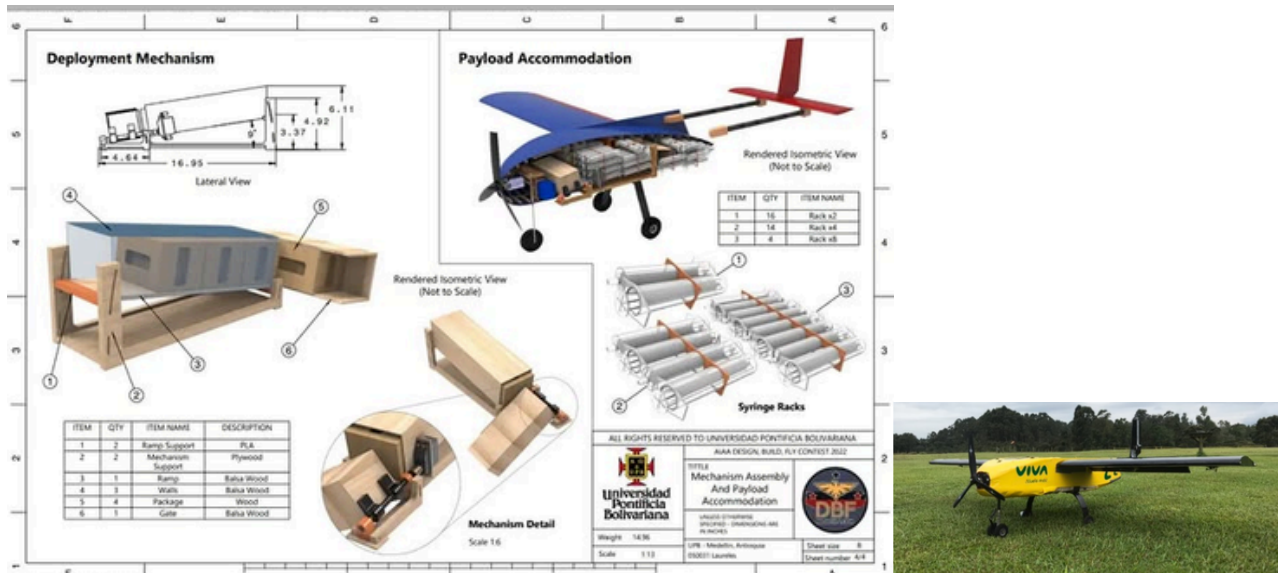


Fig. 1 — JACAMAR (DBF 2022). CAD layout of the payload bay and syringe-rack deployment mechanism, showing integration with the fuselage and electrical routing for the medical-supply mission.

One-liner: Selected 3/40 candidates to the team; sized propeller and battery and balanced the electrical budget for JACAMARÁ, a fixed-wing UAS for DBF-2022’s medical-supply mission.

Problem. Deliver medical supplies in timed missions with ground staging and precise deployment—2022 DBF theme required flying syringes and vaccine-vial packages and completing three flight missions plus a ground demo; fly-off 22–24 Apr 2022, Wichita (Textron Aviation Employees’ Flying Club).

Approach.

- Propeller/battery selection via momentum theory + motor curves; trade-study on disk loading, climb margin, and 25-ft takeoff constraint (M1–M3).
- Electrical power budget (ESC, Rx/servos, avionics) and wiring/connector standards; quick-turn battery swaps for the 5-min staging windows.
- Weight & balance updates as payload configurations changed (syringe count, vial container).

My contribution.

- Led propeller–motor–battery matching and electrical architecture; authored test cards for thrust/current/voltage sweeps; coordinated ground-staging checklist.
- Assisted with mission readiness (tech inspection wing-tip load planning; taxi/remote-deploy logic per M3).



Fig. 2 — team members and all subsystems assembled at the time of completion of the model aircraft’s construction

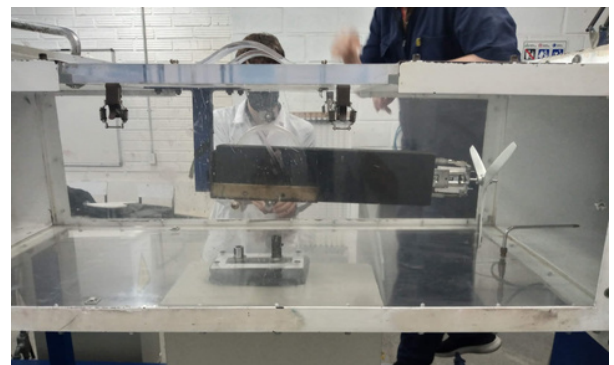


Fig. 3 — Wind tunnel tests to evaluate different propellers and their impact on energy consumption

Results. Team qualified and flew at the DBF fly-off in Wichita; UPB DBF Medellín Team is an officially recognized entrant (17th position in the world ranking).

Stack. Prop-sizing (XFOIL/JavaProp or manufacturer curves), eCalc, CAD, ground-staging fixtures, checklists.

More Evidence:

- AIAA DBF 2022 Fly-off rankings and scoring for the 2022 competition (Wichita): mission results, technical/report scores, and overall standings. ([Link](#))

CAREER PROJECTS (Course & Lab Builds) — Selected Highlights | 2021–2024



Figure 1 — Right: hand-launched glider built for the Introductory Aerospace Engineering course; left: combustion nozzle rig with flow and injection control developed for Aerospace Systems II.

Glider Design Challenge — Team Lead | 2021

- Goal. Non-powered glider, <1 kg, longest glide time.
- Design. 1.8 m span, balsa composite structure, optimized CG/dihedral; hand-launch tests.
- Outcome. 2nd place; avg. glide 3.84 s (competition format).
- Stack. CAD, hand-calc aerodynamics, workshop fabrication, field testing.

Nozzle/Combustion Rig — Controls & Instrumentation | 2022

- Goal. Build a didactic nozzle/combustion demo with staged temperature sensing and safe ignition.
- Design. Inlet plenum, combustion chamber with injector manifold, convergent–divergent nozzle; surface thermometry (LM35), gas detection (MQ), humidity probes; Arduino-based DAQ & interlocks. (High-level academic rig; no propellant formulas nor procedures included for safety.)
- Outcome. Successful cold-flow & controlled hot-fire demonstrations for lab evaluation.
- Stack. Arduino C/C++, basic instrumentation, safety interlocks, data logging.

Pilatus PC-12 ECS Mock-up — Mechatronics | 2023

- Pilatus PC-12 ECS Mock-up — Mechatronics | 2023 Goal. PLA scaled mock-up of the AC Control System with actuated valves/ducts. Design. Arduino Mega control of dampers; dry ice for visible cooling airflow; panel switches. (PC-12 uses conditioned bleed air mixed to target cabin temp; mock-up emulates flow/valving only.) Outcome. Functional demo showing modes, distribution, and temperature-mixing concepts. Stack. Arduino, 3D printing, basic control logic.

High-Lift Wing Section with Actuated Surfaces — Test Lead | 2024

Goal. Characterize $CL-\alpha$ and stall behavior with flaps/slats/spoiler actuation in a wind tunnel. Design. Arduino-driven surface deflections; multi-point force measurement; repeatable angle-setting jig. Outcome. Lift/drag curves across configurations; qualitative stall-margin improvement documented. Stack. Wind-tunnel testing, data reduction, uncertainty notes, Python plotting.

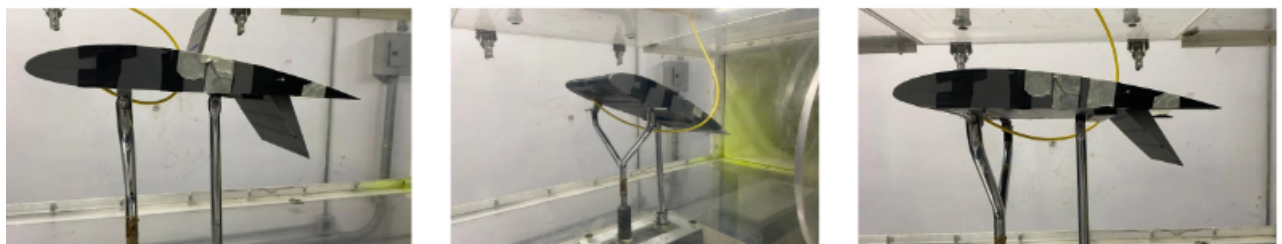


Fig. 2 – Wind-tunnel experiment of a wing with flap, slat, and spoiler actuation. Arduino-driven mechanisms allowed controlled deflection and repeatable force measurements for $CL-\alpha$ characterization.

More Evidence:

- Flight Test Video: Short clip showing the glider prototype's flight performance, highlighting aerodynamic stability and endurance optimization. ([Link](#))
- Combustion Rig Demonstration: Controlled hot-fire test video from the laboratory, illustrating temperature-sensing calibration and safe ignition sequence. ([Link](#))
- Design Documentation: Technical report including hybrid propulsion modeling, electrical power distribution diagrams, and CFD results for spray uniformity analysis. ([Link](#))

FIRST LEGO LEAGUE — Robot Design & Mechanical Lead | 2018–2019 | Tecnoacademia SENA (Risaralda, Colombia)

One-liner: Designed and built an autonomous LEGO EV3 robot to complete mission circuits; team won 1st place (Pereira regional) and 5th at nationals (Bogotá).

Problem. Build and program a LEGO robot to execute a set of field missions and present an Innovation Project on a real-world theme.

Approach.

- Concept and mechanical layout (drive, gearing, tool mounts) for reliable mission cycling.
- EV3-based coding (state machine for mission sequences; sensor-based line following). (EV3 is permitted in FLL Challenge; SPIKE Prime is the current default.) FIRST
- Innovation Project: plastic-waste recycling for flood-control retaining walls (research + poster).

My contribution.

- Led concept/mechanical design; assisted with program architecture and test plans.
- Coordinated field-run repeatability (fixture alignment, battery management, quick-swap tools).

Results. Pereira regional 1st; Nationals (Bogotá) Top-5.

Stack. LEGO MINDSTORMS EV3, color/ultrasonic sensors; EV3 Lab/Scratch; team poster.

More evidence:

- Demo video (run sequence): [LINK](#)
- High school Magazine article / press note: [LINK](#)



Fig. 1 — Modular attachment for multi-mission tasks , color/ultrasonic sensing, rigid bracing for consistent placements.



Fig. 2 —Team coordinating mission sequence on the official table; robot staged with quick-swap tools..